

# Edentulous Image Prediction using Latent Diffusion Model and Variational Autoencoder

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## 1. INTRODUCTION

Any tooth extraction results in bone loss of varying degrees [1, 2]. This leads to a residual bone volume (RBV) after healing that cannot be known *a priori* by practitioners. This inherent uncertainty directly impacts implant planning, as the final RBV dictates implant positioning, stability, and the necessity for invasive augmentation procedures.

It is important to distinguish the specific type of bone loss addressed in this study. The term *peri-implant bone loss* is often used ambiguously in the literature. It may refer to (i) a *secondary* infectious-inflammatory process around implants (peri-implantitis) [3], or (ii) a *primary* physiological remodeling process immediately following extraction. In this work, we focus exclusively on the latter: predicting the post-extraction anatomy resulting from natural physiological remodeling, independent of subsequent inflammatory diseases.

Currently, predicting this remodeling is challenging due to patient-specific variability. Standard deterministic approaches fail to capture the range of possible healing outcomes. To address this, we hypothesize that generative deep learning can model the probabilistic relationship between pre-extraction anatomy and post-extraction geometry. Specifically, we formulate this as a conditional image-to-image generation task [4]. By leveraging Latent Diffusion Models (LDMs), we aim to generate plausible edentulous sagittal slices conditioned on dentate inputs, thereby providing a tool to assess the probabilistic limits of bone healing.

## 2. RELATED WORKS

### 2.1. Variational Autoencoder (VAE)

VAEs serve as a foundational framework for perceptual compression in medical imaging [5]. By mapping high-dimensional images into a probabilistic latent space, VAEs encourage the model to learn a continuous and structured representation of anatomy rather than memorizing pixel-level data. However, a known limitation of standard VAEs is feature entanglement, where specific anatomical characteristics cannot be easily isolated. To improve interpretability, recent works have proposed architectures like the Attribute-Regularized VAE (AR-VAE) [6], which introduces regularization losses to enforce monotonic relationships between latent dimensions and continuous data attributes.

### 2.2. Latent Diffusion Models (LDM)

Generative modeling has evolved significantly with the introduction of diffusion models. While offering high mode coverage and training stability compared to GANs [7], pixel-space diffusion is computationally expensive. To address this, Rombach et al. introduced Latent Diffusion Models (LDMs) [8], which shift the diffusion process into the compressed latent space of a pre-trained VAE. This approach reduces computational complexity while retaining high-frequency spatial details. In the medical domain, diffusion models are increasingly adopted for their ability to model uncertainty [9]; by formulating generation as an iterative denoising process [10], they can approximate the distribution of possible anatomical outputs rather than forcing a single deterministic prediction.

Crucially, while original diffusion architectures were unconditional, clinical applications typically require precise control over generation. To achieve this, recent advancements utilize conditioning encoders to inject auxiliary information—such as class labels or scalar metrics—directly into the network. This is commonly transferred via cross-attention layers [11, 8] in the denoising U-Net backbone, allowing the model to guide the image synthesis toward specific target characteristics while maintaining the semantic richness of the underlying diffusion prior.

## 3. METHODOLOGY

### 3.1. General Pipeline

The strategy is first to select the VAE that provides the most suitable dentate latent representation to feed the diffusion model. This latent vector is then used by the LDM to predict the corresponding edentulous latent, which is finally decoded into the image-space by the same VAE.

### 3.2. Dataset and Metric Extraction

The study utilizes a dataset of 6011 paired sagittal slices derived from 137 CBCT examinations ( $256 \times 256$ , TIF float32). Since the images for the same patient look similar, we split our dataset into 10% test and 90% train-val with no patient overlap between the two sets. To ensure robust quantitative evaluation, an algorithm is developed to extract six morphological metrics (1 bone height and 5 widths) from each pair.

The extraction pipeline addresses the challenge of defining the alveolar crest in dentulous images, where dental structures obscure the bone limit. Consequently, the geometric

reference frame is established exclusively on the *edentulous* mask. First, a bounding box is computed around the edentulous bone to determine the ground-truth bone height ( $H$ ). This vertical span is then divided into equidistant intervals to define five specific depths for width measurement. Crucially, these vertical coordinates are projected onto the paired *dentulous* slice. This methodology ensures that the five dentulous bone widths ( $W_{1-5}$ ) are measured at spatially identical anatomical levels to their edentulous counterparts.

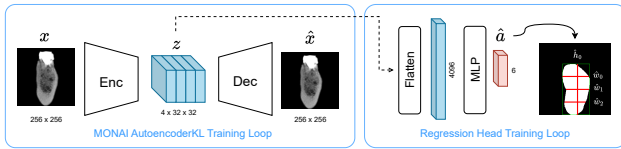
### 3.3. VAE Architectures and Training Ablations

We trained three baseline VAE models—one on dentate images only, one on edentulous images only, and one jointly on both—to obtain a first full-dataset baseline validation and assess convergence behaviour. Preliminary experiments showed that adversarial losses caused rapid overfitting and unstable training dynamics, with a non-informative KL term.

Therefore, adversarial losses were removed for the final VAE training. Accordingly, the VAE loss is defined as follows, where  $\lambda_{\text{KL}}$  and  $\lambda_{\text{perc}}$  respectively weight the KL divergence and perceptual loss terms,  $\mathcal{S}$  denotes the set of selected perceptual layers, and  $\psi_{\ell}(\cdot)$  the corresponding feature extractor [12, 13].

$$\mathcal{L}_{\text{VAE}} = \|x - \hat{x}\|_1 + \lambda_{\text{KL}} D_{\text{KL}}(q(z|x) \parallel \mathcal{N}(0, I)) + \lambda_{\text{perc}} \sum_{\ell \in \mathcal{S}} \|\psi_{\ell}(x) - \psi_{\ell}(\hat{x})\|_2^2. \quad (1)$$

Since a dentate image is encoded and subsequently provided to the LDM, we focus on training the dentate-only and joint (both) VAEs using a  $(4 \times 32 \times 32)$  latent space, with a 4-level encoder featuring  $32 - 64 - 128 - 128$  as channel dimensions, as illustrated in Fig. 1.



**Fig. 1.** VAE and Regression Head Training Loops

We also trained a dentate-only VAE using our adaptation of the AR-VAE framework. Leveraging MONAI’s AutoEncoderKL [14], which outputs a spatial latent tensor  $z^{(i)} \in \mathbb{R}^{C \times H \times W}$ , we expanded the latent dimensionality to 10 channels to accommodate the regularization of six edentulous attributes (1 height and 5 widths). Following the AR-loss formulation [6], we adapt the distance term to our latent representation by defining it as the spatial mean of channel  $k$ :

$$D_r(i, j) = \bar{z}_k^{(i)} - \bar{z}_k^{(j)}, \quad \bar{z}_k^{(i)} = \frac{1}{HW} \sum_{u,v} z_{k,u,v}^{(i)} \quad (2)$$

However, t-SNE [15] and UMAP [16] analysis showed that while the baseline VAEs formed clear patient-specific

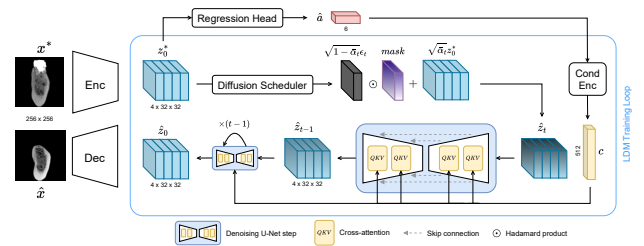
clusters, the AR-VAE produced more mixed embeddings, suggesting that the expanded 10-channel latent space may dilute the available information. These experiments also confirmed that the AR-VAE failed to regularize its latent space: the KL term diverged rapidly.

### 3.4. Edentulous Metrics Regression Head

Since the AR-VAE did not provide a stable or usable latent structure, we instead opted to regress the edentulous metrics directly from the dentate latent vector encoded by a frozen VAE. As shown in Fig. 1, the spatial latent tensor  $z$  is flattened and fed into a Multi-Layer Perceptron (MLP) with a 4096-unit hidden layer to predict the six anatomical metrics  $\hat{a}$ . This decoupled approach allows us to map the dentate latent codes to the edentulous measurements without attempting to regularize the latent space during VAE training. This creates a mapping between the abstract latent manifold and the physical geometric constraints required for LDM conditioning.

### 3.5. Latent Diffusion Models

We employ a Latent Diffusion Model (LDM) operating in the latent space of the *VAE-both* architecture (chosen for its robust encoding/decoding of both domains). The model learns to generate an edentulous latent  $\hat{z}_0$  by reversing a  $T=1000$ -step diffusion process with a U-Net denoiser, as illustrated in Fig. 2.



**Fig. 2.** LDM Training Loop

A standard formulation would initialize the reverse process from pure Gaussian noise and use both the dentate latent  $z_0^*$  and the metrics  $\hat{a}$  as conditioning signals. In our setting, this led to excessive stochasticity and poor preservation of the basal bone structure. We therefore adopt a partial-diffusion initialization: instead of using  $z_0^*$  as an explicit conditioning input, we use it as the starting point of the diffusion trajectory and add spatially selective noise to obtain the initial state  $\hat{z}_T$  for denoising.

#### 3.5.1. Conditioning Mechanism

To ensure the generated anatomy respects the patient’s specific morphology, the U-Net backbone is conditioned on the

six extracted bone metrics. We implement a Conditioning Encoder (CondEnc) adapted from [17]. Following this formulation, the scalar metrics vector  $\hat{a} \in R^6$  is first processed by a dropout layer ( $p = 0.1$ ), which randomly nullifies the conditioning vector during training to improve robustness and prevent over-reliance on the conditioning signal, before being projected to a 512-dimensional embedding via learnable weights  $W_1$  and  $W_2$  with a SiLU activation:

$$c = W_2 \cdot \text{SiLU}(W_1 \cdot \text{Dropout}(\hat{a})) \quad (3)$$

The resulting embedding  $c \in R^{512}$  is injected into the U-Net’s intermediate layers via cross-attention mechanisms, guiding the denoising process towards the target geometry.

### 3.5.2. Spatially-Selective Gradient Noising

To preserve patient-specific basal structures while allowing variability in the ridge region, we apply a spatially varying noise mask during the forward diffusion of the dentate latent  $z_0^*$  (Fig. 6). Let  $\epsilon \sim \mathcal{N}(0, I)$  and  $\text{mask} \in [0, 1]^{H \times W}$  be a vertical gradient mask (low values in the basal area, higher values towards the dental area), broadcast across latent channels. Using the MONAI diffusion scheduler coefficients  $\bar{\alpha}_t$  [10, 14], we construct the noised latent at step  $t$  as:

$$\hat{z}_t = \sqrt{\bar{\alpha}_t} z_0^* + \sqrt{1 - \bar{\alpha}_t} (\epsilon \odot \text{mask}). \quad (4)$$

The denoiser is then trained with the standard diffusion objective by predicting the injected noise, where  $\epsilon$  denotes the Gaussian noise realization used to form  $\hat{z}_t$  and  $\theta$  represents the parameters of both U-Net and CondEnc:

$$\mathcal{L}_{LDM} = E_{t, \epsilon} [\|\epsilon - \hat{\epsilon}_\theta(\hat{z}_t, \hat{a}, t)\|^2]. \quad (5)$$

This setup encourages the network to modify primarily the high-noise (crest) region while maintaining structural consistency in low-noise (basal) areas.

## 4. RESULTS AND DISCUSSION

### 4.1. VAE Performance and Latent Space Analysis

**Table 1.** Comparison of refined VAEs on test set.

| ID   | MSE↓                      | SSIM↑        | PSNR↑       | Perc.↓        | KL↓          |
|------|---------------------------|--------------|-------------|---------------|--------------|
| 1-d  | 3.72e <sup>-4</sup>       | 0.980        | 34.7        | 0.0071        | <b>453</b>   |
| 1-ed | 4.33e <sup>-4</sup>       | 0.982        | 33.5        | 0.0065        | <b>421</b>   |
| 2-d  | 4.57e <sup>-4</sup>       | 0.977        | 33.9        | 0.0080        | 0.133        |
| 2-ed | 6.35e <sup>-4</sup>       | 0.980        | 32.2        | 0.0081        | 0.117        |
| 3-d  | <b>3.10e<sup>-4</sup></b> | <b>0.986</b> | <b>35.5</b> | <b>0.0060</b> | <b>0.106</b> |
| 3-ed | <b>3.43e<sup>-4</sup></b> | <b>0.989</b> | <b>34.9</b> | <b>0.0045</b> | <b>0.075</b> |

**Model IDs:** 1 = AR-VAE-dente, 2 = VAE-dente, 3 = VAE-both.

**Test set IDs:** d = dentate test set, ed = edentate test set.

Table 1 summarizes reconstruction and regularization metrics on the held-out test set. VAE-dente achieves the

best reconstruction on dentate slices, whereas the joint VAE-both provides the most consistent performance across both domains and attains the best overall trade-off (lowest MSE and perceptual loss, highest SSIM/PSNR, and lowest KL). The t-SNE projection further indicates that both VAE-dente and VAE-both yield well-clustered patient-specific latents, with dentate and edentulous embeddings from the same patient lying in close proximity (Fig. 5). In contrast, AR-VAE exhibits unstable behaviour, with KL values orders of magnitude larger than the other models, indicating a poor match to the unit-Gaussian prior and an uninformative latent regularization.

### 4.2. Latent Space Metric Regression

**Table 2.** Comparison of regression head config. on test set.

| Encoder   | Target scale   | MAE↓        | MSE↓        | $R^2$ ↑      |
|-----------|----------------|-------------|-------------|--------------|
| VAE-both  | normalized     | 5.11        | 48.2        | 0.235        |
| VAE-dente | normalized     | <b>4.94</b> | <b>46.8</b> | <b>0.250</b> |
| VAE-both  | non-normalized | 6.70        | 79.6        | -0.169       |
| VAE-dente | non-normalized | 6.93        | 73.5        | -0.245       |

**Table 3.** Attribute-wise details for the best regression head config. (VAE-dente + normalized targets).

| Attribute | Target (Test set) |          | Regression metrics |      |              |
|-----------|-------------------|----------|--------------------|------|--------------|
|           | $\mu$             | $\sigma$ | MAE↓               | MSE↓ | $R^2$ ↑      |
| height_0  | 174               | 18.5     | 9.18               | 129  | 0.451        |
| width_0   | 62.7              | 6.06     | 3.33               | 19.7 | -0.303       |
| width_1   | 67.3              | 8.25     | 4.66               | 41.6 | 0.010        |
| width_2   | 71.9              | 10.4     | 3.81               | 23.0 | 0.365        |
| width_3   | 75.2              | 11.2     | 3.94               | 24.1 | <b>0.698</b> |
| width_4   | 62.9              | 12.1     | 4.73               | 43.7 | 0.281        |

Table 2 shows that target normalization is critical for stable regression: without normalization, performance collapses with negative  $R^2$ . With normalized targets, the dentate-only encoder yields the best results ( $\text{MAE} = 4.94$ ,  $R^2 = 0.25$ ), slightly outperforming VAE-both. Table 3 further highlights strong attribute-dependent behaviour: height\_0 and width\_3 are moderately predictable ( $R^2 = 0.45$  and  $0.70$ , respectively), while width\_0 remains poorly captured ( $R^2 < 0$ ). Overall, these results suggest that only a subset of the post-extraction geometry is reliably encoded in dentate latent representations.

### 4.3. LDM Error Analysis and Visual Samples

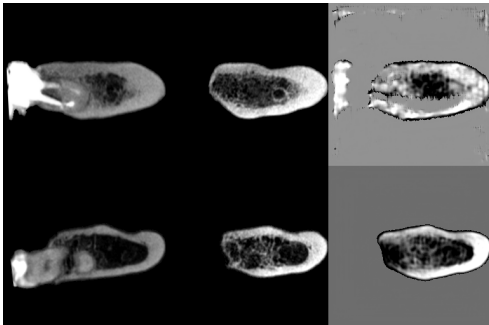
Fig. 3 reports the residual  $\hat{e} - e_{\text{ed}}$  as a function of the ground-truth remodeling ratio  $e_{\text{ed}}/e_{\text{den}}$ . For mild remodeling (ratios close to 1), residuals concentrate around zero for most attributes, indicating that the LDM produces anatomically consistent outputs when the post-extraction geometry remains



**Fig. 3.** Residual analysis of LDM-predicted edentulous metrics. Each subplot corresponds to one attribute (*height\_0*, *width\_0*–4). The x-axis shows the ground-truth remodeling ratio  $e_{ed}/e_{den}$  (edentulous over dentate), and the y-axis shows the prediction residual  $\hat{e} - e_{ed}$ , where  $\hat{e}$  is the metric extracted from the LDM-predicted edentulous slice and  $e_{ed}$  is the ground-truth edentulous metric.

close to the dentate configuration. As remodeling becomes more pronounced (ratio decreases), the dispersion increases and structured biases emerge, suggesting that the most severe cases remain challenging.

For *height\_0*, residuals become mostly positive at low ratios, suggesting an underestimation of vertical resorption (predicted edentulous height remains too large). Width errors are more heterogeneous: intermediate depths stay relatively stable, whereas deeper levels show higher variance with occasional large outliers. These outliers mostly correspond to rare catastrophic generation failures (Fig. 4, failure case), where the whole slice reconstruction is corrupted rather than a simple metric mismatch. Importantly, this failure mode was not observed when sampling from pure Gaussian noise (outputs were less consistent but remained visually coherent), and appears specifically after introducing the spatial gradient noising, indicating that the partial-diffusion initialization can sometimes destabilize the denoising trajectory.



**Fig. 4.** Qualitative LDM samples (success vs. failure): dentate GT, edentulous GT, and LDM-predicted edentulous (top: failure case; bottom: successful case).

Fig. 4 complements this quantitative analysis with two

representative samples. In the successful case (bottom), the predicted edentulous slice closely matches the ground-truth edentulous anatomy while preserving patient-specific basal structures visible in the dentate input. In contrast, the failure case (top) shows clear artifacts and geometric inconsistencies in the predicted ridge, resulting in large metric deviations. These qualitative observations align with the residual plots, highlighting that, while the proposed spatially selective noising stabilizes most predictions, robustness remains limited for the most challenging remodeling patterns.

## 5. CONCLUSION

This study investigates generative deep learning for patient-specific post-extraction bone remodeling prediction, a key challenge for implant planning. We propose a conditional LDM operating in the latent space of a refined VAE. Results show that the joint VAE (trained on both domains without adversarial loss) provides the best reconstruction trade-off, while dentate latents only partially encode the information needed to predict edentulous geometry.

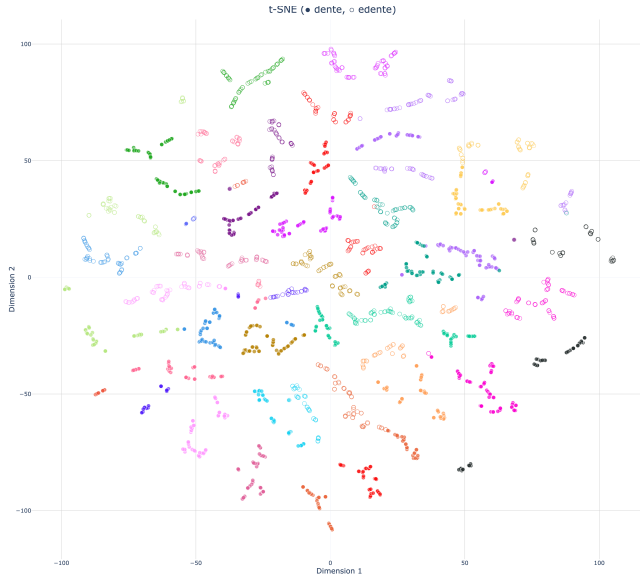
The regression analysis indicates that while certain dimensions (e.g., height) are predictable, the intrinsic biological variability of healing remains difficult to infer solely from pre-extraction anatomy.

Methodologically, the introduction of spatially selective gradient noising successfully addressed the challenge of preserving patient-specific basal bone. However, this comes at the cost of stability, and severe remodeling cases occasionally lead to geometric inconsistencies. Future work should focus on integrating non-anatomical clinical metadata (e.g., age, time since extraction) to better constrain the biological uncertainty and improve the robustness of the partial-diffusion initialization to prevent reconstruction failures.

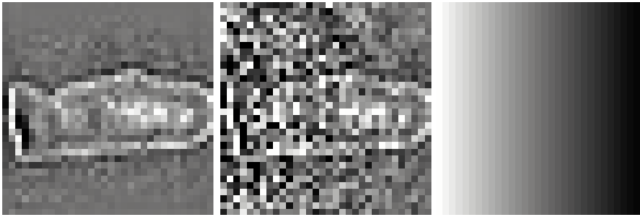
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## Appendix



**Fig. 5.** t-SNE projection of VAE latent representations encoded via VAE-both. Colors indicate patient identity; filled markers correspond to dentate slices and hollow markers to edentulous slices.



**Fig. 6.** Example of scheduler-based latent noising with spatial masking. Left: clean latent channel. Middle: noised latent channel at step  $t$  produced by the diffusion scheduler. Right: vertical gradient mask used to modulate the injected noise.